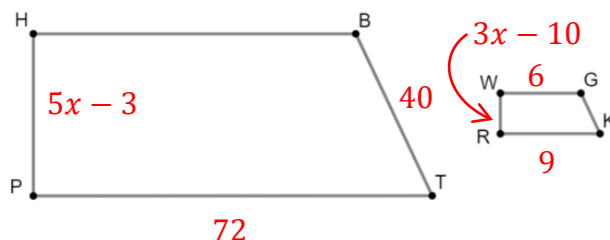


Geometry
Unit 6
Lesson 13 Practice

Name Answer Key
Date _____
Period _____

Wendy, a party planner, is designing a banner for a retirement party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.

Wendy knows that $PT = 72$ inches (in.),
 $HP = (5x - 3)$ in., $WG = 6$ in., $WR = (3x - 10)$ in.,
 $BT = 40$ in., and $RK = 9$ in.



1. What scale factor will map $WGKR$ to $HBTP$?

$$\frac{72}{9} = 8$$

2. What is the perimeter of $HBTP$, the main entrance banner?

$$HB = 6 \cdot 8 = 48 \text{ in}$$

$$HP = 5 \left(\frac{77}{19} \right) - 3 \approx 17.26 \text{ in}$$

$$P = 72 + 40 + 48 + 17.26 = 177.26 \text{ in}$$

$$5x - 3 = 8(3x - 10)$$

$$5x - 3 = 24x - 80$$

$$19x = 77$$

$$x = \frac{77}{19} \approx 4.0526$$

3. What is the perimeter of $WGKR$, the photo booth banner?

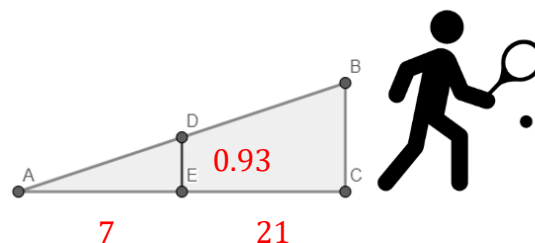
$$GK = \frac{40}{8} = 5$$

$$WR = 3 \left(\frac{77}{19} \right) - 10 \approx 2.16 \text{ in}$$

$$P = 9 + 5 + 6 + 2.16 = 22.16 \text{ in}$$

Answers to #2 and #3 may vary slightly due to rounding of the value of x .

In the diagram, $\triangle ADE \sim \triangle ABC$, $AE = 7$ meters (m),
 $DE = 0.93$ m, and $EC = 21$ m. Serena is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 7 m away from the base of the net.



4. What scale factor will map $\triangle ABC$ to $\triangle ADE$?

$$\frac{7}{28} = \frac{1}{4}$$

5. What is the height, represented by $\overline{BC}$, of the tennis racket?

$$\frac{7}{28} = \frac{0.93}{BC}$$

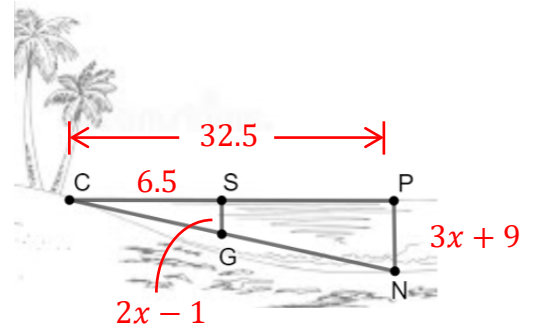
$$7 \cdot BC = (0.93)(28)$$

$$BC = 3.72 \text{ m}$$

A lake front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 6.5$ feet (ft), $SG = (2x - 1)$ ft,

$PN = (3x + 9)$ ft, and $CP = 32.5$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?

$$\frac{6.5}{32.5} = 0.2 \text{ or } \frac{1}{5}$$

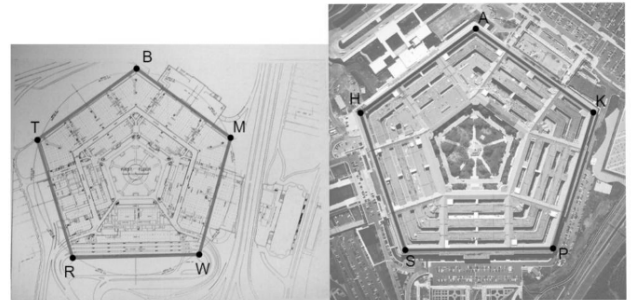
7. What is the depth of the longer support post, represented by $\overline{PN}$?

$$\begin{array}{lcl} \frac{2x - 1}{3x + 9} = \frac{1}{5} & 5(2x - 1) = 3x + 9 & PN = 3x + 9 \\ 10x - 5 = 3x + 9 & 7x = 14 \rightarrow x = 2 & PN = 3(2) + 9 \\ & & PN = 15 \text{ ft} \end{array}$$

8. What is the depth of the shorter support post, represented by $\overline{SG}$?

$$\begin{array}{l} SG = 2x - 1 \\ SG = 2(2) - 1 \\ SG = 3 \text{ ft} \end{array}$$

Cesar wants to create a scale model of the Pentagon, where all sides are equal, for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{6}$ of the scale model Pentagon, $AKPSH$. $TB = (0.5x + 5)$ centimeters (cm) and $HA = (5x + 10)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?

$$\begin{array}{lcl} \frac{0.5x + 5}{5x + 10} = \frac{1}{6} & 6(0.5x + 5) = 5x + 10 & P = 5 \cdot [(0.5)(10) + 5] \\ 3x + 30 = 5x + 10 & 2x = 20 & P = 50 \text{ cm} \\ x = 10 & & \end{array}$$

10. What is the perimeter of the scale model Pentagon, $AKPSH$?

$$\begin{array}{l} P = 5 \cdot [(5)(10) + 10] \\ P = 300 \text{ cm} \end{array}$$